# K-COMP: Retrieval-Augmented Medical Domain Question Answering With Knowledge-Injected Compressor

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## Problem Statement

**Challenges in closed-domain question-answering**

- Closed-domain QA requires domain-specific knowledge, which pre-training alone often fails to capture.
- Retrieval-augmented methods rely heavily on retrievers and corpora, raising concerns about their reliability.

To address this, we train **K-COMP** to compress documents by ***focusing on the jargon present in the question***, ultimately generating concise context that fulfills the knowledge required to answer it.

## Prerequisites

📚 Retrieval Corpus for retrieval-augmentation

To ensure coverage of both general and domain knowledge, we adopt the MedCorp (Xiong et al.).

🔎 Retriever for relevant context selection

We employ an embedding-based k-NN search model (i.e., encoder models).

✂️ Data processing for fine-tuning

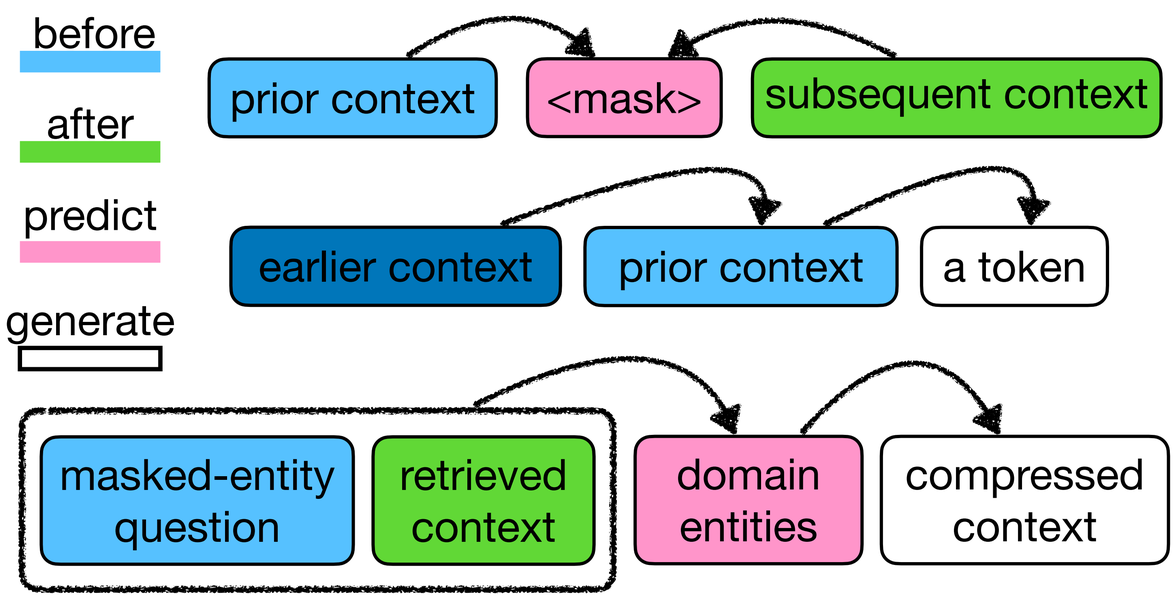
Ground-truth summaries are synthesized using gpt-4o-mini, and entities are recognized with spaCy-based tool.

## Approach

### ① Causal Knowledge Injection

**Leveraging model-specific attributes**

- Masked language models (e.g., BERT) excel at predicting masked spans by referencing attention scores from tokens located ***both before and after.***
- Causal models using auto-regressive language modeling (e.g., GPT) have an advantage in causally generating entire texts ***by referring to preceding tokens.***




### ② Preliminary

**Masked-entity question & Entities description**

$q$ = What are the symptoms of down syndrome ?

$q_m$ = What are the <ent> of <ent>?

$d_1$ = symptom: {{description}}<eod>

$d_2$ = Down syndrome: {{description}}<eod>

**Passages & Domain-knowledge**

$p_i$ = retrieved passage

$e_i$ = entity included in the question (e.g., symptom)

$s$ = ground-truth summary

$\mathscr{E} = \{e_1 \ldots e_n\} \quad \mathscr{D} = \{d_1 \ldots d_n\} \quad \mathscr{P} = \{p_1 \ldots p_5\}$

### ③ Training

$$P_\theta(\mathscr{E}, \mathscr{D}, s \mid q_m \oplus \mathscr{P}) = P_\theta(\mathscr{E}, \mathscr{D} \mid q_m \oplus \mathscr{P}) \times P_\theta(s \mid \mathscr{E}, \mathscr{D}, q_m \oplus \mathscr{P})$$

$$\therefore L(\theta) = -\mathbb{E}(\log P_\theta(\mathscr{E}, \mathscr{D}, s \mid q_m \oplus \mathscr{P}))$$

## Main Results

Models of all sizes place greater trust in K-COMP's concise descriptions than in other baselines. 🚀

| | General-purpose LLMs | | | | | | | | Medical-purpose LLMs | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Llama-3-8B | | Llama-3-70B | | Mixtral-8x7B | | GPT-4o[7] | | MedAlpaca-13B | | Meditron-70B | |
| | BertScore | UniEval | BertScore | UniEval | BertScore | UniEval | BertScore | UniEval | BertScore | UniEval | BertScore | UniEval |
| MedQuAD | | | | | | | | | | | | |
| **Without compressor** | | | | | | | | | | | | |
| Top-1 document | 75.65 | 53.44 | 79.44 | 57.47 | 58.68 | 44.30 | 84.45 | 60.24 | 75.05 | 36.89 | 76.38 | 52.74 |
| Top-5 documents | 76.20 | 52.45 | 79.95 | 55.25 | 64.17 | 47.47 | 83.08 | **64.98** | 17.57 | 10.82 | 73.14 | 51.88 |
| **With compressor** | | | | | | | | | | | | |
| RECOMP | 73.27 | 55.74 | 79.75 | **63.57** | 71.58 | 58.78 | 85.20 | 60.92 | 82.52 | 39.15 | 76.41 | 56.14 |
| LLMLingua | 76.45 | 54.65 | 78.78 | 57.00 | 59.79 | 45.34 | 84.46 | 55.04 | 83.10 | 44.17 | 74.74 | 54.18 |
| FineTune | 74.27 | 54.89 | 79.97 | 59.67 | 71.44 | 56.26 | 83.47 | 62.40 | 82.25 | 39.40 | 76.14 | 55.97 |
| K-COMP | **83.79** | **64.05** | **82.06** | 62.91 | **80.64** | **63.07** | **85.49** | 63.16 | **85.75** | **49.39** | **78.88** | **58.95** |
| MASH-QA | | | | | | | | | | | | |
| **Without compressor** | | | | | | | | | | | | |
| Top-1 document | 77.87 | 54.08 | 82.65 | 56.48 | 72.63 | 51.88 | 83.39 | 60.66 | 76.95 | 43.41 | 80.45 | 55.31 |
| Top-5 documents | 77.59 | 52.94 | 82.07 | 57.20 | 73.09 | 53.18 | 83.12 | **64.81** | 31.91 | 20.07 | 79.31 | 55.85 |
| **With compressor** | | | | | | | | | | | | |
| RECOMP | 80.20 | 58.80 | 82.27 | 60.50 | **82.12** | 62.35 | 83.82 | 62.22 | 78.51 | 34.83 | 81.54 | 60.09 |
| LLMLingua | 79.74 | 54.37 | 83.25 | 56.83 | 76.71 | 58.32 | 83.37 | 56.73 | 80.41 | 48.45 | 80.78 | 58.55 |
| FineTune | 77.30 | 55.21 | 82.90 | 60.66 | 78.74 | 60.62 | 83.88 | 62.82 | 79.96 | 44.02 | 81.70 | 60.06 |
| K-COMP | **83.45** | **61.45** | **83.64** | **61.40** | 81.49 | **62.91** | **83.92** | 64.06 | **81.95** | **51.76** | **82.67** | **61.85** |
| BioASQ | | | | | | | | | | | | |
| **Without compressor** | | | | | | | | | | | | |
| Top-1 document | 82.53 | 56.03 | 81.40 | 59.58 | 82.23 | 62.07 | 87.13 | 62.47 | 76.86 | 40.06 | 83.43 | 57.55 |
| Top-5 documents | 81.44 | 61.33 | 80.91 | 61.09 | 83.54 | 65.53 | 86.81 | **66.15** | 20.93 | 12.41 | 83.63 | 59.52 |
| **With compressor** | | | | | | | | | | | | |
| RECOMP | 83.89 | 59.80 | 85.45 | 59.65 | **86.85** | 62.82 | 87.87 | 60.62 | 78.07 | 30.68 | 84.77 | 57.58 |
| LLMLingua | 81.70 | 53.62 | 86.53 | 61.53 | 82.54 | 62.24 | 86.85 | 51.11 | 81.19 | 42.97 | 81.51 | 55.26 |
| FineTune | 82.26 | 60.89 | 86.17 | 62.86 | 85.48 | 64.01 | **87.97** | 63.99 | 78.67 | 41.49 | 85.05 | 61.20 |
| K-COMP | **86.61** | **62.28** | **87.03** | **62.89** | 86.52 | **66.31** | 87.77 | 64.38 | **84.88** | **45.47** | **86.70** | **62.90** |

🧐 **Further analysis in the paper**

- Is K-COMP effective in unseen environments?
- Case study on contexts compressed by K-COMP
- Evaluation using a state-of-the-art LLM
- Inference speed analysis, and more

## Entity Description is Effective ?

| | Llama-3-8B | | Llama-3-70B | | MedAlpaca-13B | | Meditron-70B | |
|---|---|---|---|---|---|---|---|---|
| | BertScore | UniEval | BertScore | UniEval | BertScore | UniEval | BertScore | UniEval |
| MedQuAD | | | | | | | | |
| K-COMP | **83.79** | **64.05** | **82.06** | **62.91** | **85.75** | **49.39** | 78.88 | 58.95 |
| −*Prior* | 73.93 | 54.47 | 80.06 | 60.03 | 84.84 | 48.23 | **79.80** | **59.00** |
| FineTune | 74.27 | 54.89 | 79.97 | 59.67 | 82.25 | 39.40 | 76.14 | 55.97 |
| MASH-QA | | | | | | | | |
| K-COMP | **83.45** | **61.45** | **83.64** | **61.40** | 81.95 | 51.76 | **82.67** | **61.85** |
| −*Prior* | 78.63 | 57.80 | 83.16 | 60.88 | **82.18** | **52.02** | 81.93 | 60.42 |
| FineTune | 77.30 | 55.21 | 82.90 | 60.66 | 79.96 | 44.02 | 81.70 | 60.06 |
| BioASQ | | | | | | | | |
| K-COMP | **86.61** | **62.28** | **87.03** | **62.89** | **84.88** | 45.47 | **86.70** | **62.90** |
| −*Prior* | 80.75 | 55.06 | 86.22 | 62.40 | 84.53 | **46.67** | 85.23 | 59.73 |
| FineTune | 82.26 | 60.89 | 86.17 | 62.86 | 78.67 | 41.49 | 85.05 | 61.20 |

## Generated Sample

Q: Are there any adults who shouldn't get the meningococcal vaccine ?

**K-COMP:**

### Entity
people: Plurality of persons considered as a whole
meningococcal vaccine: Vaccine against meningococcal disease

### Summary
An auto-regressively generated summary based on domain knowledge relevant to the question.